

Home Plant Automation

Hardware Purchase List & Full Wiring / Connection Diagram

Scope 10-pot Dendrobium orchid setup · 5 Thai hybrid varieties (Bangkok Peach, Pop Eye, Yellow Stripe, Caesar Pink, Thong Deang)

Location Parakandeniya, Kadawatha, Sri Lanka

Controller ESP32, 7-channel relay · misting + fertigation + fungicide + auto-fill

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This document consolidates every hardware item identified for the test-bench build — what's already purchased, what's still needed, and where to buy it — followed by the complete electrical and plumbing connection diagram matching the exact GPIO pin assignments in the live firmware (`firmware/esp32-unit/src/Config.h`).

On prices: every link below is a real, previously-verified listing from the supplier named — none are fabricated. Live price lookup was unavailable at the time this document was generated (a browser-tool outage), so the price column reads “See listing” throughout. Confirm the exact price, stock, and spec (voltage / thread size / NC vs NO) on the linked page before ordering — marketplace listings are not always precise about these.

Hardware List

Organized by what's already covered, what's critical, and what's optional. All items sourced from real Sri Lankan marketplaces — Daraz.lk, tronic.lk, duino.lk, alphontronic.lk, BNS Hardware, PE Plus, and ThingsInNet.

A. Already have — no action needed

Item	Store	Price
OLED display (SSD1306, I2C)	Daraz.lk	Owned
DHT22 temperature/humidity sensor	Daraz.lk	Owned
Rain sensor module (digital out)	Daraz.lk	Owned
SSR relay board (AC-only — being replaced, see Critical)	Daraz.lk	Owned
Diaphragm pump, 12V, 4.5 L/min	Daraz.lk	Owned
Solenoid valve, 12V DC, NC (x2)	Daraz.lk	Owned
12V 5A SMPS power supply	Daraz.lk	Owned
LM2596 buck converter (12V→5V)	Daraz.lk	Owned
DC barrel jack	Daraz.lk	Owned
ESP32 dev board + prototyping shield	Daraz.lk	Owned
Misting kit (x2)	Daraz.lk	Owned

B. Control board — optional integrated alternative

A single pre-built board that folds the ESP32 + 4 relays + power regulation + OLED into one PCB.

Item	Store	Price	Notes
ESP32 Universal IoT Dev Kit – Actuating	ThingsInNet	See listing	Optional single-board alternative: ESP-WROOM-32D-N8, 4 opto-isolated relays, OLED, buzzer, 2 status LEDs, 2 hardware-debounced dry-contact ports (ideal for the water-level float switches), 8–24V input. Covers 4 of the 7 relay channels; still needs a 3-channel add-on relay module for fungicide pump/valve + inlet valve, and a firmware pin remap in Config.h.

C. Critical — blocks the bench from working correctly

The SSR relay board already owned is AC-only and cannot reliably switch the 12V DC pumps/valves.

Item	Store	Price	Notes
12VDC 4-Way 4-Channel Relay Module (DB9)	Tronic.lk	See listing	Replaces the AC-only OMRON SSR — needed to reliably switch the 12V DC pumps/valves. Get two for the full 7-channel build.
12V 4-Channel Relay Module w/ Optocoupler LED	Duino.lk	See listing	Alternative source for the same relay-board swap.

D. Recommended — reliability, cheap insurance

Item	Store	Price	Notes
Panel Mount Fuse Holder 5x20mm	Tronic.lk	See listing	Goes right after the SMPS, before anything else on the 12V bus.
Ceramic Fuse 5A (25x6mm)	Tronic.lk	See listing	Matches the SMPS's 5A rating.
White Box Enclosure, waterproof, 196x90x60mm	Tronic.lk	See listing	Houses the ESP32 + relay board(s) + buck converter — this now lives somewhere humid.
Non-contact Tank Liquid Level Sensor, 12–24V	Tronic.lk	See listing	Clamps outside the tank wall, no moving parts. Need two: low + full.

Also needed, new in this revision — mechanical float/limit switch (overflow cutoff). Mounted above the full sensor as a hard backstop; no specific verified listing found yet, so no link is given here rather than fabricate one. Search tronic.lk / duino.lk / Daraz for “float switch water level” or “limit switch water tank” and confirm voltage/contact type before ordering.

E. Dosing pumps

Genuine 12V peristaltic pumps, from Sri Lankan stock.

Item	Store	Price	Notes
Kamoer 12V DC Mini Peristaltic Pump NKP-DC-S06B	Alphatronic.lk	See listing	Genuine 12V peristaltic dosing pump, Sri Lankan stock.
Kamoer Peristaltic Dosing Pump, 420ml/min, 12V/24V	Duino.lk	See listing	Alternative source for the same class of pump.

F. Solenoid valves

Item	Store	Price	Notes
Electric Solenoid Valve 1/2" 12VDC NC	Tronic.lk	See listing	

Item	Store	Price	Notes
Electric Solenoid Valve, L-shape angle, 0.5" 12VDC NC	Tronic.lk	See listing	
1/2" Plastic Solenoid Valve 12V DC (pressure-free)	Nilambaara Electronics	See listing	
12V Solenoid Water Valve, 3/4" / 1"	Alphatronic.lk	See listing	
12V Electric Solenoid Valve	LK-Tronics	See listing	

G. Check valve

Item	Store	Price	Notes
Non Return Valve — Era brand	BNS Hardware	See listing	Proper 1/2" pipe-thread check valve (not a generic aquarium-tubing part) — non-negotiable on the feed line to stop fertilizer siphoning back into the shared manifold.

H. Tanks & drums

Item	Store	Price	Notes
Empty HDPE Plastic Blue Drum, 50L	Daraz.lk	See listing	Main Water Tank — open top, mesh lid only (never sealed). The mains supply is chlorinated, so this tank has to stand open to air for the chlorine to off-gas; the firmware now holds misting for 24h after every auto-fill to give it time. A 500–1000L sealed tank is unnecessary at this scale and would only make that dechlorination wait slower and messier to manage.
Empty HDPE Plastic Blue Drum, 30L	Daraz.lk	See listing	Stock tank, for fertilizer/fungicide concentrate.
Phoenix Plastic Bucket with Lid, 100L	Daraz.lk	See listing	Mixing tank.

I. Basic prototyping supplies

Item	Store	Price	Notes
Solderless Breadboard + Jumper Wire Kit	Tronic.lk	See listing	

Item	Store	Price	Notes
5.08mm 2-Pin Screw Terminal Block	Tronic.lk	See listing	

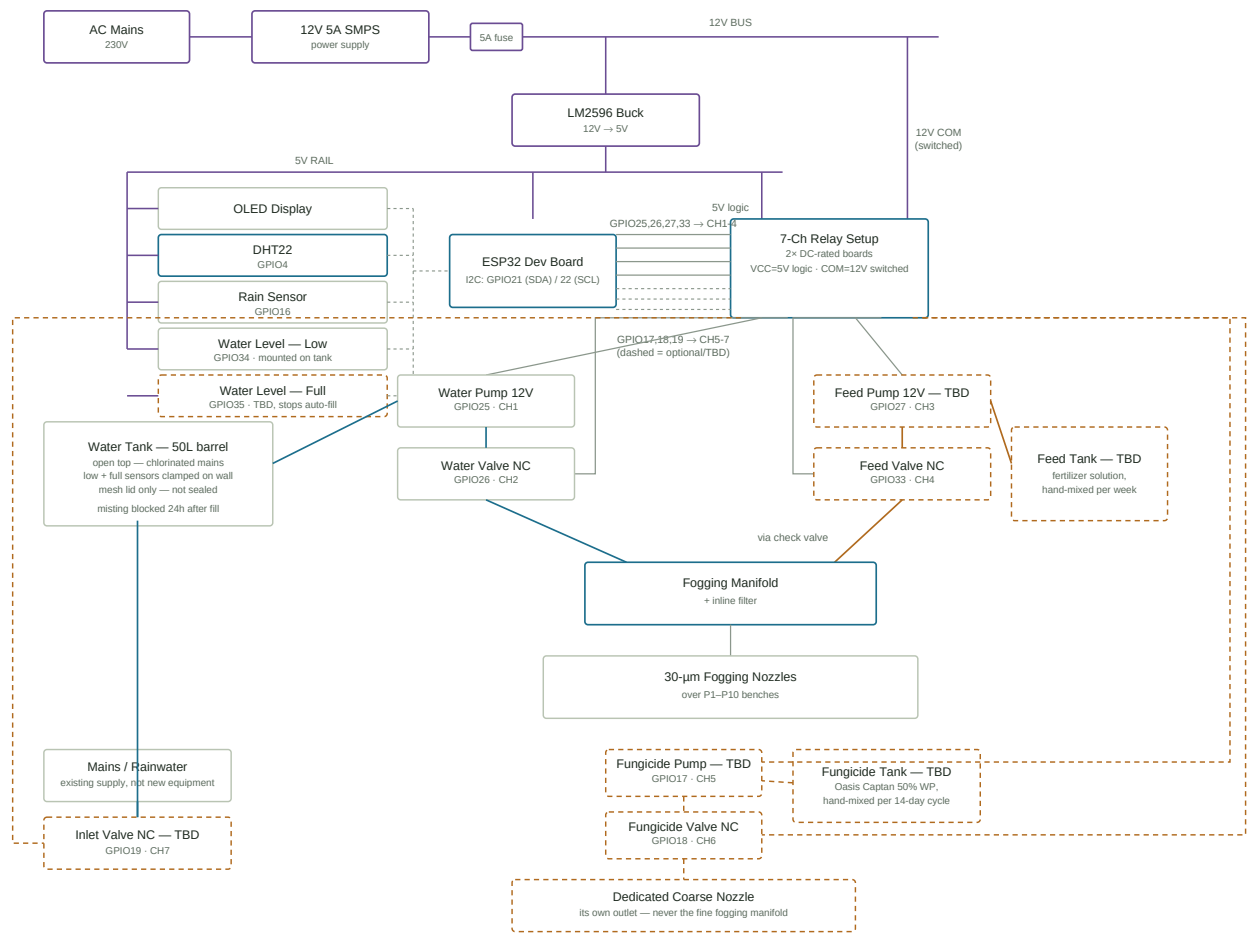
J. Optional — precision dosing measurement

Hold off until the basic mist/feed loop is proven reliable — this is the Phase 3 upgrade, not required for the current bench.

Item	Store	Price	Notes
TDS Water Conductivity Sensor	Tronic.lk	See listing	Phase-3 upgrade — EC-based dosing precision, not required for the current 10-pot bench.
Load Cell 1Kg	Tronic.lk	See listing	Weight-based stock-tank measurement.
HX711 Dual-channel 24-bit AD Module	Tronic.lk	See listing	Load-cell amplifier — pairs with the load cell above.

Full Wiring & Connection Diagram

Purple = power distribution (mains → 12V → 5V, and the relay's separate switched 12V). Grey = GPIO control signal. Blue = plain water. Amber = fertilizer feed. Dashed amber = not yet ordered (the fungicide line deliberately never joins the shared fogging manifold — it sprays through its own coarse, dedicated nozzle instead, so wettable-powder residue can't clog the fine 30-µm nozzles).



Power Budget

Worst realistic case is one pump + its own valve energized at once — firmware interlocks never run two lines simultaneously, and never open a valve without its pump. The inlet valve is the one exception: auto-fill can overlap a mist cycle, adding a fixed ~0.5A rather than compounding the pump/valve worst case. Either way, well inside the SMPS's 5A rating.

Device	Rail	Typical draw	Notes
ESP32 dev board	5V	80–500 mA	Spikes during Wi-Fi transmit bursts
OLED display	5V	~20 mA	
DHT22	5V	~1.5 mA	
Rain sensor module	5V	~15 mA	
Water level sensor (each)	5V	~20 mA	Non-contact type — two of these, low + full
Relay module (per energized coil)	5V logic	~70–80 mA	At most 2 coils on at once by design
Diaphragm pump 12V 4.5L/min	12V switched	~0.8–1.5 A	Largest single load — size wiring for this
Solenoid valve 12V (each)	12V switched	~0.3–0.5 A	Draws continuously while held open — up to 3, plus the inlet
Fungicide dosing pump (peristaltic)	12V switched	~0.1–0.3 A	Much lighter load than the diaphragm pumps
Worst case (diaphragm pump + its valve, + auto-fill overlap)	12V	~2.5 A	Against a 5A supply — comfortable headroom

Wire gauge: at least 18 AWG for anything carrying pump current (12V runs to pumps/valves, SMPS-to-relay-COM run); 22–24 AWG is fine for logic/signal wiring (GPIO lines, I2C, sensor VCC).

Pin-by-Pin Table

Matches `firmware/esp32-unit/src/Config.h` exactly — if you rewire differently, update that file too, or the firmware will drive the wrong pins.

Component pin	ESP32 pin	Notes
OLED SDA	GPI021	I2C, address 0x3C
OLED SCL	GPI022	
DHT22 data	GPI04	Add a 10kΩ pull-up to 3.3V if your module lacks one
Rain sensor digital out	GPI016	Verify polarity with a wet-finger test
Water level — low, signal	GPI034	Input-only pin. Raw “liquid present” reading; firmware inverts it to decide “needs a fill”
Water level — full, signal	GPI035	Input-only pin. Needed for auto-fill
Water level — overflow limit switch	GPI036	Input-only pin, mounted above the full sensor. GPIO34-39 have no internal pull resistor — wire an external one. Always active, forces the inlet shut regardless of <code>autofill_enabled</code>
Relay IN1 (water pump)	GPI025	
Relay IN2 (water valve)	GPI026	
Relay IN3 (feed pump)	GPI027	Reserved — feed pump not yet ordered
Relay IN4 (feed valve)	GPI033	
Relay IN5 (fungicide pump)	GPI017	Opt-in — wire only if <code>fungicide_automated</code> is on
Relay IN6 (fungicide valve)	GPI018	Feeds the dedicated coarse nozzle — never the fine fogging manifold
Relay IN7 (water inlet valve)	GPI019	Opt-in — wire only if <code>autofill_enabled</code> is on
Relay VCC (logic side)	5V rail	Same rail as ESP32 — do not connect this to the 12V bus
Relay COM (switched side)	12V bus	This is what actually powers the pumps/valves through the relay contacts
Boot button (Wi-Fi reset)	GPI00	Hold 3s at power-on to re-pair
All GND (ESP32, relay logic, sensors)	common ground	Every ground must tie together, or relays never see a valid trigger level

Dechlorination Hold

The mains supply is chlorinated, so the Water Tank is deliberately an open, vented 50L barrel rather than a sealed 500–1000L tank — chlorine off-gasses into open air over time, but only if the tank isn't airtight. The firmware enforces a wait on top of that: every time the auto-fill inlet valve closes (full sensor tripped, or the safety timeout), it records that moment and blocks the misting line from drawing on that tank until the dechlorination hold has passed — 24 hours by default. This applies even to a forced `mist_now` request; the guard is not skippable from the app.

Now a per-unit setting, not a fixed firmware constant. “Dechlorination hold (days)” and “Pump flow rate (L/min)” are editable directly in the dashboard's Schedule form (Water Tank section) and sync to the device like any other setting — no reflash needed to retune them. Same goes for fertilizer/fungicide dosing: the form now takes a mix ratio (mL concentrate per liter of water) and a batch water volume, and computes the actual dose automatically.

24 hours is a starting point, not a verified number. It's a common rule-of-thumb dwell time for free-chlorine off-gassing by standing, but the right value depends on the actual chlorine dosing used by the local water authority (NWSDB). Confirm this locally and adjust the hold in the app if needed.

Constant / behavior	Value	Notes
Dechlorination hold	24h default	Misting hold after each fill — editable per-unit in the app; not independently verified, confirm locally
AUTOFILL_MAX_SECONDS	3 min (180)	Resized down from 10 min for the 50L barrel — a stuck full-sensor at the old timeout would overflow a tank this size. Still a fixed firmware safety constant, not app-editable
Overflow limit switch	GPIO36	Hard mechanical cutoff above the full sensor — always active, forces the inlet shut and raises an “overflow” event regardless of auto-fill or app settings
Persistence	NVS	Last-fill timestamp survives a reboot, so a power blip doesn't quietly reset the hold early
Status surfaced as	“dechlorinating” / “overflow”	Shown on the OLED and in cloud status instead of a silent “idle” while either condition is active

Tank & Plumbing Routing

Connection	Detail
Water Tank → Water Pump	Short suction hose, strainer/mesh at the tank outlet
Water Pump → Water Valve → Manifold	Straight line, no branching — the daily plain-misting path
Feed Tank → Feed Pump	Same as the water side, once the feed pump is ordered
Feed Pump → Feed Valve → check valve → Manifold	The check valve is non-negotiable — without it, fertilizer can siphon back into the shared manifold
Water level sensors (low + full)	Both clamp to the outside of the Water Tank wall at different heights — non-contact
Water Tank cover	Loose mesh lid only, never sealed — chlorine needs to off-gas into open air. Keeps debris and mosquitoes out without trapping chlorine in
Overflow limit switch	Mounted on the tank wall above the full sensor — a last-resort mechanical cutoff if the non-contact full sensor sticks or fails. Trips regardless of whether auto-fill is enabled
Manifold → Nozzles	Inline filter right before the nozzles — last line of defense against clogging
Fungicide Tank → Fungicide Pump → Fungicide Valve → dedicated coarse nozzle	Its own plumbing run, start to finish — never joins the water/feed manifold
Mains/Rainwater → Inlet Valve → Water Tank	Opens when the low sensor clears, closes on the full sensor or the firmware's timeout safety cutoff

Before You Power It On

Nine things worth checking with a multimeter first — all cheap and all easy to get backwards on a first build.

- **Relay VCC vs COM are two different rails.**

VCC (logic) is 5V from the buck converter; COM (switched) is 12V straight from the bus. Wiring the pump's 12V into the logic VCC pin will kill the relay board.

- **Common ground, always.**

ESP32 GND, relay logic GND, and the 12V supply's ground all need to be tied together — otherwise the GPIO signal has no shared reference and the relay won't trigger reliably.

- **Confirm active-low vs active-high on the relay board before flashing.**

Most cheap boards energize on a LOW signal. Config.h's RELAY_ACTIVE_LOW flag must match your actual board, or every relay will be inverted.

- **Fuse goes right after the SMPS, before anything else.**

It protects the whole 12V bus — anywhere else, part of the circuit is unprotected.

- **Power up logic before actuators.**

Get the ESP32 booting and the OLED showing readings first, pump/valve wires disconnected. Only connect 12V actuator wiring once the relay clicks on command.

- **Physically double-check the fungicide line never tees into the water/feed manifold.**

This is a plumbing mistake, not a wiring one — firmware can't detect it. Wrong, and the first automated dose runs wetttable-powder residue through the 30- μ m nozzles.

- **Test the auto-fill timeout before trusting it unattended.**

Cover the full sensor so it never triggers, and confirm the inlet valve closes on its own after AUTOFILL_MAX_SECONDS (3 min default, resized for the 50L barrel).

- **Don't seal the Water Tank lid.**

The mains supply is chlorinated, and the tank relies on standing open to air for the chlorine to off-gas before the firmware will mist with it (DECHLORINATE_SECONDS, 24h default). A sealed or airtight lid defeats this — use loose mesh instead.

- **The 24h dechlorination hold resets on every fill, not just the first one.**

Each auto-fill event adds fresh chlorinated water, so the firmware restarts the wait every time the inlet valve closes — misting can go quiet for a day after any top-up, not just the initial fill. Confirm the actual dwell your water authority's chlorine level needs and adjust DECHLORINATE_SECONDS if 24h is wrong for your supply.